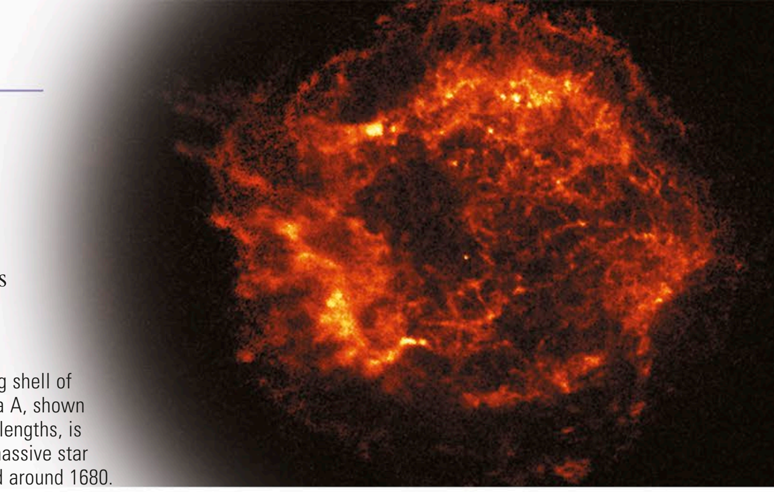
STELLAR END POINTS

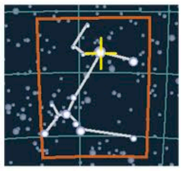
Stars end their lives in a variety of ways, but many are difficult or impossible to observe. It is thought that unobserved dead stars contribute significantly to the Milky Way's mysterious missing mass (see pp.226–229). Often, black holes and small white dwarfs can be observed only by the effect they have on nearby objects, and neutron stars are visible only in gamma-ray wavelengths. However, some stellar end points and their remnants, such as supernovae, are among the galaxy's most spectacular sights.



STAR REMNANT
A rapidly expanding shell of hot gas, Cassiopeia A, shown here in X-ray wavelengths, is the remnant of a massive star that died unnoticed around 1680.

WHITE DWARF

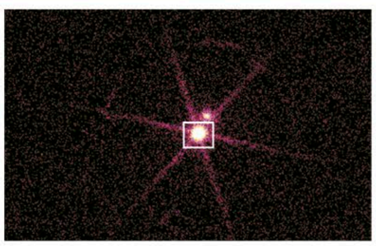
Sirius B



CATALOG NUMBER HD 48915 B
DISTANCE FROM SUN 8.6 light-years
MAGNITUDE 8.5

CANIS MAJOR

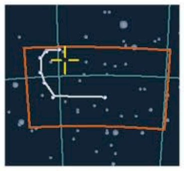
This was the first white dwarf to be discovered. First observed in 1862, it was found to be a stellar remnant when its spectrum was analyzed in 1915. Although Sirius A, its companion, is the brightest star in the sky, Sirius B appears brighter in X-ray images (such as the one below). Sirius B's diameter is only 90 percent that of Earth, but since its mass is equal to that of the Sun, its gravity is 400,000 times that on Earth.



CLOSE COMPANIONS

NEUTRON STAR

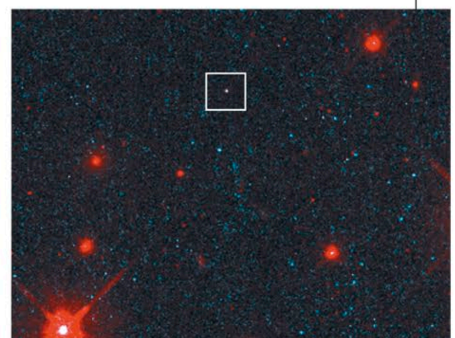
RX J1856.5-3754



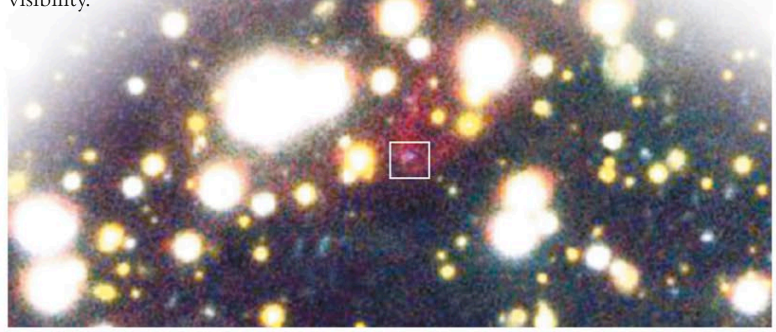
CATALOG NUMBER 1ES 1853-37.9
DISTANCE FROM SUN 200–400 light-years
MAGNITUDE 26

CORONA AUSTRALIS

This lone star is one of the closest known neutron stars to Earth. Discussions are ongoing as to its true distance, but estimates vary from 200 to 400 light-years. There is also much speculation about its age. Some astronomers believe it is an old neutron star emitting X-rays because it is accreting material onto its surface from the surrounding interstellar medium. Others believe it is a young neutron star emitting X-rays as it cools. It is possible that it formed about 1 million years ago, when a massive star in a close binary system exploded. It is traveling through the interstellar medium at about 240,000 mph (390,000 kph). RX J1856.5-3754 is moving away from a group of young stars in the constellation of Scorpius. Also moving away from this group of stars is the ultra-hot blue star now known as Zeta (ζ) Ophiuchi. It is possible that RX J1856.5-3754 is the remnant of Zeta Ophiuchi's original binary companion. As the closest neutron star, it is being extensively studied, but its diminutive size makes it difficult for astronomers to obtain conclusive results. Estimates of the diameter of RX J1856.5-3754 vary from 6 miles (10 km) to 20 miles (30 km). This puts it very close to the theoretical limit of how small a neutron star can be, challenging some models of their internal structure. Its X-ray emissions suggest it has a surface temperature of around 1,000,000°F (600,000°C). Its visual magnitude of only 26 means that this star is 100 million times fainter than an object on the limit of naked-eye visibility.



RARE VIEWS
Taken in 1997, a Hubble image (above) offered astronomers an unusual glimpse of a neutron star in visible light. The star's movement through the interstellar medium has produced a cone-shaped nebula, visible in a later image (below).



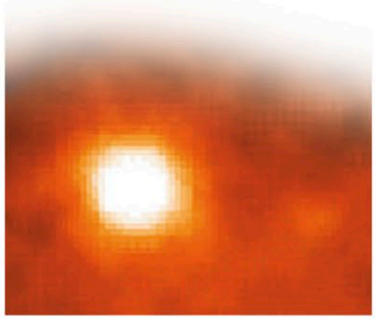
NEUTRON STAR

Geminga Pulsar

CATALOG NUMBER SN 437
DISTANCE FROM SUN 500 light-years
MAGNITUDE 25.5

GEMINI

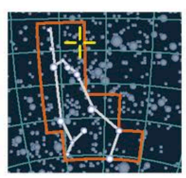
Discovered in 1972, the Geminga Pulsar, a pulsating neutron star, is the second-brightest source of high-energy gamma rays known in the Milky Way. Its name is a contraction of "Gemini gamma-ray source"; it is also an expression, in the Milanese dialect, meaning "It's not there," because only recently has this object been observed in wavelengths other than gamma rays. Variations in the pulsar's period of luminosity (see pp.280–281) have suggested that it may have a companion planet, but they may also be due to irregularities in the star's rotation. Geminga is believed to be the remnant of a supernova that took place about 300,000 years earlier in the star's life. It is traveling through space at almost 15,000 mph (25,000 kph), at the head of a shock wave 2 billion miles (3.2 billion km) long.



GAMMA RAY SOURCE
The Geminga Pulsar shines bright in an image taken through a gamma-ray telescope. Gamma-ray photons are blocked from Earth's surface by the atmosphere.

WHITE DWARF

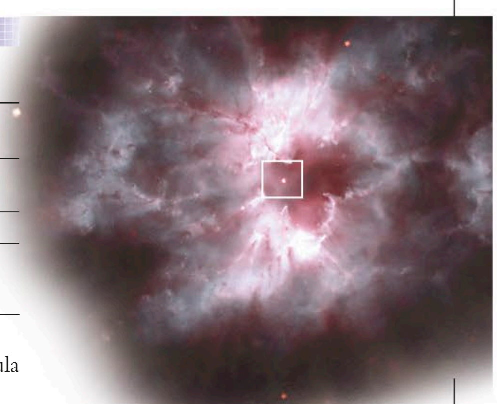
NGC 2440 nucleus



CATALOG NUMBER HD 62166
DISTANCE FROM SUN 3,600 light-years
MAGNITUDE 11

PUPPIS

The central star of the planetary nebula NGC 2440 has one of the highest surface temperatures of all known white dwarfs. This stellar remnant has a surface temperature of around 360,000°F (200,000°C)—40 times hotter than that of the Sun. This also makes it intrinsically very bright, with a luminosity more than 250 times that of the Sun. The complex structure of the surrounding nebula has led some astronomers to believe that there have been periodic ejections of material from the dying central star. The structure of the nebula also suggests that the material was ejected in various directions during each episode.



INNER LIGHT
Energy from the extremely hot surface of NGC 2440's central white dwarf makes this beautiful and delicate-looking planetary nebula fluoresce.